

Research Questions

Primary Question

How do we architect autonomous procurement agents that make reliable, cost-effective purchasing decisions independently while maintaining transparency, learning from outcomes, and preventing catastrophic errors?

Sub-Questions

1. What autonomous agent framework enables scheduled monitoring, multi-step workflows, and learning loops?

- Framework comparison: LangGraph (known, good for workflows), AutoGen (multi-agent conversations), CrewAI (role-based), n8n (rapid prototyping)
- Decision factors: scheduling capability, memory integration, learning system support
- Recommendation direction: LangGraph (familiar, proven for complex workflows)

2. How should memory systems track supplier reliability over time?

- Requirements: supplier history, pricing trends, delivery performance, past decision outcomes
- Memory patterns: short-term (current transaction), mid-term (supplier scoring window), long-term (historical trends)
- Storage considerations: Zep for session memory, structured DB for supplier profiles
- Scoring evolution: how does one bad experience update reliability scores without eliminating good suppliers?

3. What decision logic prevents bad purchasing decisions?

- Multi-factor evaluation: price vs delivery speed vs supplier reliability
- EV-style calculations: purchase cost vs stockout prevention value
- Threshold logic: when to choose expensive-but-fast vs cheap-but-slow
- Constraint checking: budget limits, approved supplier lists, maximum autonomous spend amounts

4. How do autonomous agents handle edge cases?

- Supplier out of stock: failover to next supplier, escalate if all unavailable
- Delivery delays: proactive monitoring, early warning to human
- Price spikes: flag unusual pricing, require human approval above threshold
- Duplicate invoices: pattern matching against recent POs
- Budget constraints: hard limits vs soft warnings

5. What safeguards prevent catastrophically bad decisions?

- Financial limits: maximum autonomous spend per transaction
- Human-in-loop: approval required for high-value or unusual decisions
- Audit trail: every decision logged with reasoning
- Rollback capability: how to reverse bad decisions
- Bias detection: fairness checks in supplier selection

- Privacy: sensitive business data handling protocols

6. Should this use multi-agent or single-agent architecture?

- Multi-agent option: Monitor Agent, Evaluator Agent, Decision Agent, Reconciliation Agent, Learning Agent
- Single-agent option: One agent with different operational modes
- Trade-offs: modularity vs simplicity, independent optimization vs coordination overhead
- Recommendation: Start single-agent for MVP, evolve to multi-agent if complexity demands